Approval Document: Automatic Storage and Retrieval System (ASRS)

A. <u>Summary:</u>

1. Overview of ASRS

An Automated Storage and Retrieval System (ASRS) is an advanced, computer-controlled solution designed to automate the storage, retrieval, and tracking of inventory within a warehouse or distribution centre. It significantly improves operational efficiency, space utilization, and accuracy by minimizing manual intervention and streamlining material handling processes.

2. Current Storage Challenges and Cost Impact

Due to increasing production volumes, we have leased external storage facilities to accommodate Finished Goods (FG) solar module pallets. Currently, the cost of leasing these storage spaces is approximately **Rs. 75 lakh to Rs. 1 crore per month**, which represents a substantial expense to the company.

Additionally, reliance on manual processes for storing and retrieving inventory has led to several operational inefficiencies:

1. **Time-Consuming FIFO (First In, First Out) Process**
    - Manually identifying and retrieving FG pallets in the correct FIFO sequence requires excessive time and effort.
    - Delays in material movement impact overall supply chain efficiency.
2. **Order Mismatch and Dispatch Errors**
    - Due to the large-scale operations, there are instances where incorrect material is shipped to customers.
    - This results in return shipments, customer dissatisfaction, and potential financial losses.
3. **Inadequate Tracking of Finished Goods in WMS**
    - The current Warehouse Management System (WMS) lacks the capability to pinpoint the exact location of stored pallets.
    - Operators must manually locate pallets, increasing the risk of errors and inefficiencies.
4. **Heavy Dependence on Manual Labor**
    - High reliance on workers for storage and retrieval increases the probability of human errors.

- Workforce constraints can lead to bottlenecks, particularly during peak demand periods.

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3. ASRS as a Strategic Solution

To address these challenges, we propose a **one-time investment** in an **Automated Storage and Retrieval System (ASRS)** that will significantly enhance warehouse efficiency while reducing recurring storage costs.

Key Features of the Proposed ASRS

- **Capacity:** The system will accommodate **8,928 pallet positions** for storing different types of solar module pallets, including **M10, M10 R, G12, and G12 R modules**.
- **Seamless Integration with SAP WMS:**
  - The ASRS will be fully integrated with the **Warehouse Management System (WMS) in SAP**, ensuring real-time tracking and location mapping of inventory.
  - As Finished Goods (FGs) arrive, their location will be automatically recorded in the system for efficient retrieval.
- **Optimized FIFO Execution:** The system will automate the FIFO process, ensuring that older stock is dispatched first without delays.
- **Minimized Errors & Mismatches:** Computerized tracking and automated retrieval will **eliminate order mismatches**, improving customer satisfaction.
- **Reduced Manpower Dependency:** Automation will **decrease reliance on manual labour**, lowering the risk of errors while enhancing overall operational speed and accuracy.

4. Expected Benefits of ASRS Implementation

- **Cost Savings:** Eliminating leased storage space will save **Rs. 75 lakhs to Rs. 1 crore per month**, resulting in long-term financial benefits.
- **Increased Efficiency:** Automated retrieval and real-time inventory tracking will significantly **reduce material movement time** and improve dispatch accuracy.
- **Better Space Utilization:** High-density storage will optimize warehouse capacity, allowing us to store more pallets in a compact area.
- **Enhanced Inventory Control:** Integration with SAP WMS will provide complete **visibility and traceability** of Finished Goods at all times.
- **Improved Order Fulfilment:** Faster and more accurate retrieval of FG pallets will **enhance customer satisfaction** and **reduce return shipments** due to incorrect deliveries.

B. Cost Analysis

1. Total Project Cost

Parameters	Quantity	Rate	Sub Total	GST	Total
ASRS			**30,60,00,000**	**5,50,80,000**	**36,10,80,000.00**
Racking system	8928	14,278	12,74,73,984	2,29,45,317.12	15,04,19,301.12
Cladding for high bay	50	2,40,000	1,20,00,000	21,60,000.00	1,41,60,000.00
Double deep Stacker crane	20	27,65,000	5,53,00,000	99,54,000.00	6,52,54,000.00
Bottom Rail with aisle accessories	4	33,62,696	1,34,50,784	24,21,141.12	1,58,71,925.12
Conveyor system	50	2,98,000	1,49,00,000	26,82,000.00	1,75,82,000.00
Critical spares	2	14,00,000	28,00,000	5,04,000.00	33,04,000.00
Metal pallets	8928	9,000	8,03,52,000	1,44,63,360.00	9,48,15,360.00
Discount			-2,76,768	-49,818.24	-3,26,586.24
Civil Cost			**7,57,37,500**	**1,36,32,750**	**8,93,70,250**
Earthworks and soiling			1,35,00,000	24,30,000	1,59,30,000.00
PCC works			45,00,000	8,10,000	53,10,000.00
RCC works			3,00,00,000	54,00,000	3,54,00,000.00
Shuttering			12,50,000	2,25,000	14,75,000.00
TMT bards Fe 500			2,10,00,000	37,80,000	2,47,80,000.00
Masonary works			6,00,000	1,08,000	7,08,000.00
Plastering			4,87,500	87,750	5,75,250.00
VDF Flooring			7,20,000	1,29,600	8,49,600.00
J BOLTS installation			16,00,000	2,88,000	18,88,000.00
Baseplate Grouting			8,80,000	1,58,400	10,38,400.00
Miscellenous			12,00,000	2,16,000	14,16,000.00
Road			**1,43,26,335**	**25,78,740**	**1,69,05,075**
Dock levellers	**8**	**4,29,544**	**34,36,352**	**6,18,543.36**	**83,84,699**
Contingency	**5%**				**2,37,87,001**
Pre - operating expenses					**54,12,517**
IDC					**3,40,26,072**
Finance cost					**26,94,828**
Total					**54,16,60,442**

Total Cost in First Year: ₹ 54.17 Cr

2. Pre- operating Expense:

Sr. No.	Position	Number	Monthly cost	Monthly total	No. of months	Total cost
1	Site Engineer	1	40,000	40,000.00	14	5,60,000.00
2	Travelling expense	1	10,000	10,000.00	14	1,40,000.00
3	Admin expenses during implementation	1	10,000	10,000.00	14	1,40,000.00
4	Consultant charges	1	0	5,00,000.00	1	5,00,000.00
5	Project insurance	1.00%		39,32,517		39,32,517.00
6	Miscellaneous Expenses	1	10,000	10,000.00	14	1,40,000.00
	Total					**54,12,517**

C. Project Design:

1. ASRS system planning:

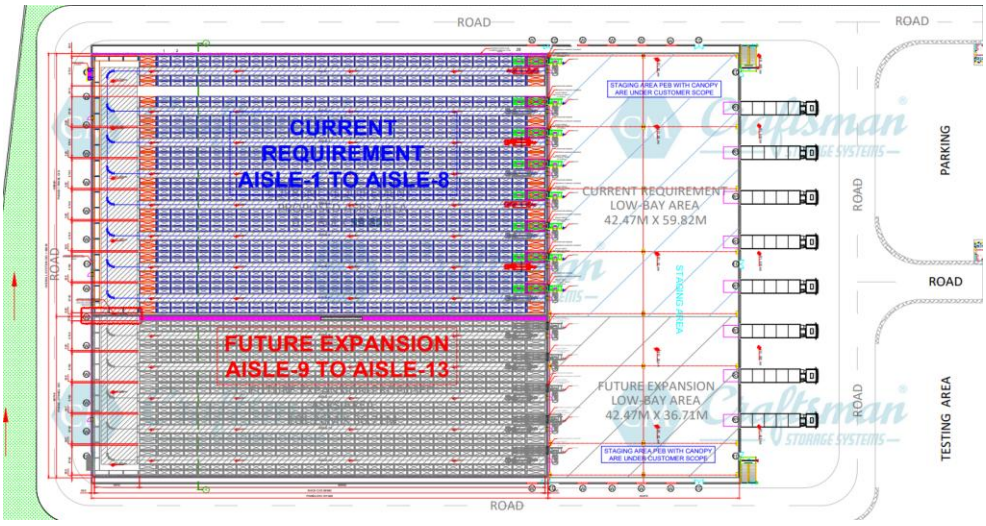


Image 1: Top view

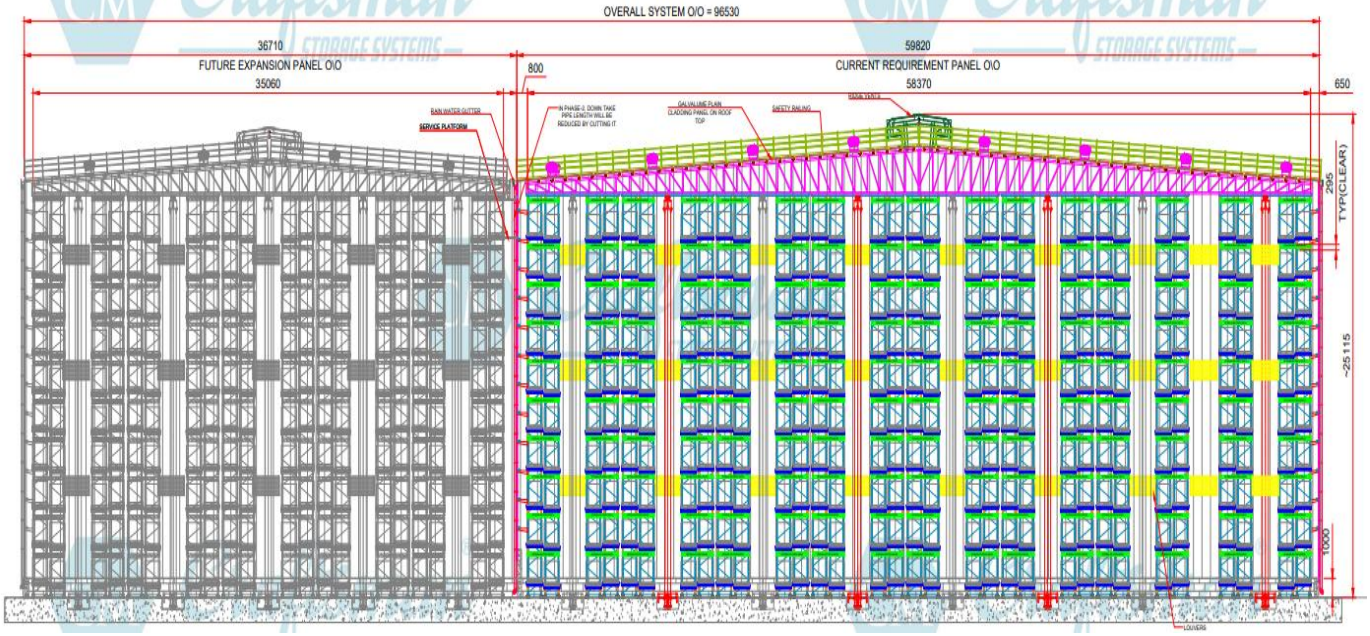
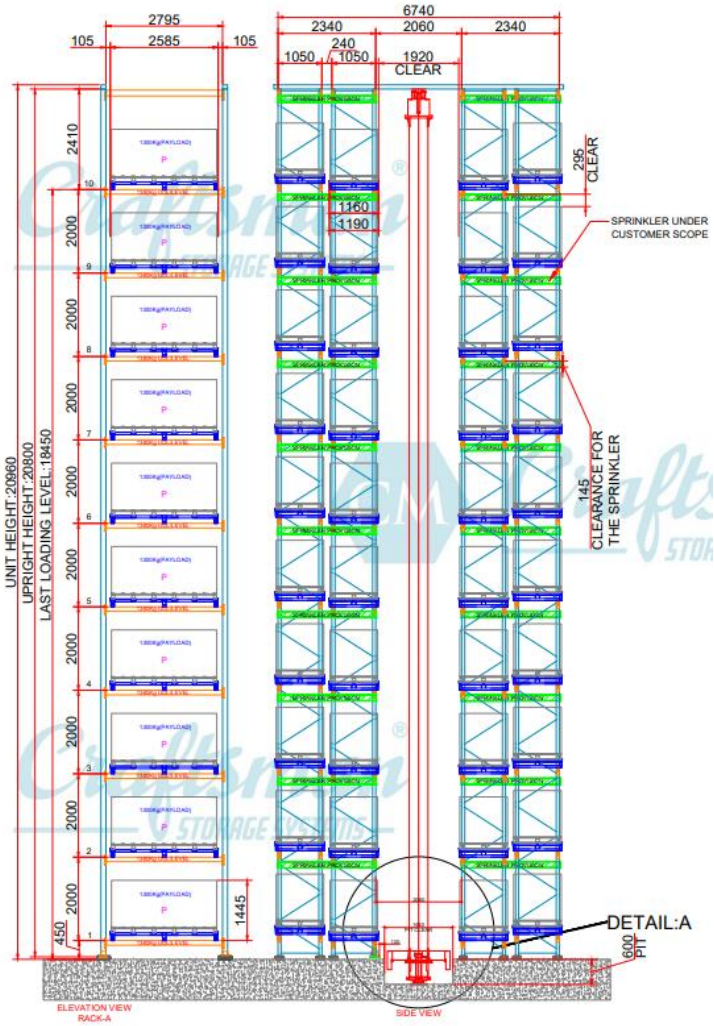


Image 2: Side view

2. ASRS racking design:



D. Project Assumptions:

Assumption List				
Sr. No.	**Parameters**	**Value**	**UOM**	**Remarks**
A	**Project**			
1	Project life	15	years	
2	Number of pallet position	8,928	#	Based on current planning
3	Rent charge per pallet per day	67.83	Rs./day/pallet	In reference to Radiant and Kothur warehouse rentals
4	Term of rent agreement	7	years	
5	Storage duration of each pallet	1	days	Material will be outbound within these days
6	Inventory Turnover	-	times	
7	Start of Project Execution	01-Oct-24		
8	End of Project Execution	30-Nov-25		
9	Land			As per the land lease rental of HSTL <> PEPPL Gas yard and Kothur warehouse rentals
a	*Total land required*	2,48,396	sq. ft.	As per the diagram

b	*Land lease rental*	16	Rs./sq. ft./month	
c	*Security deposit on land*	3	months	
d	*Term of land rental*	7	years	
e	Escalation			
	Yr 1	0%	%	
	Yr 2	0%	%	
	Yr 3	5%	%	
	Yr 4	5%	%	
	Yr 5	5%	%	
	Yr 6	5%	%	
	Yr 7	5%	%	
B	**Financial**			
1	Total project cost	54.17	Rs. Crore	
a	Debt	54.17	Rs. Crore	
b	Equity	0	Rs. Crore	100% equity funded
2	WACC	7.11%	%	9.5% cost of debt and 20% cost of equity.
3	Tax Rate	25.17%	%	
4	Depreciation term	7.5	years	

E. Project Returns:

Returns		
Parameters	**Value**	**UOM**
IRR	13.10%	%
NPV	66.40	crore
Payback period	5.16	years
ROCE	35.58%	%

Detailed cash flow is given in Annexure 1.

F. Reason for Proposal:

1. Existing Warehouse Setup:

When compared to the current **Radiant** and **Kothur** warehouse, the key details of the existing storage facility are as follows:

Radiant:

- Total Available Space: 71,858 sq. ft.
- Maximum Storage Capacity: 2,590 pallets
- Average Storage Duration per Pallet: 1 days
- Lease Cost: Rs. 22 per sq. ft. per month
- Security Deposit: Equivalent to 3 months' rent
- Lease Tenure: 1 years

Kothur:

- Total Available Space: 2,61,597 sq. ft.
- Maximum Storage Capacity: 9,355 pallets
- Average Storage Duration per Pallet: 1 days
- Lease Cost: Rs. 23 per sq. ft. per month

- Security Deposit: Equivalent to 3 months' rent
- Lease Tenure: 3 years

- Cost Breakdown of the Warehouse

Expense Parameter	Radiant Warehouse (Rs. Cr.)	Kothur Warehouse (Rs. Cr.)
Lease cost		
Lease Rent	2.24	8.52
Security Deposit	0.56	2.13
Total Lease Cost	**2.80**	**10.65**
Add: Operational cost		
Electricity	**0.03**	**0.03**
Manpower	**0.27**	**0.62**
IT Costs	**0.02**	**0.02**
Machinery & Equipment	**0.55**	**1.62**
Transport Cost	**0.21**	**6.20**
Total Cost	**3.87**	**19.14**

Based on this, the storage cost per pallet amounts to average cost of **Rs. 58.98 per pallet per day**.

2. Limitations of the Current Warehouse Setup
   - **High Operational Costs Without Additional Benefits:**
     The existing setup incurs significant expenses, including lease rent, manpower, and machinery depreciation, without providing any operational efficiencies.
   - **Manual Effort for Order Fulfilment:**
     Retrieving specific customer materials involves a **multi-step process**, where the team must first **identify the required pallet, manually extract it from storage, and load it onto trucks** for dispatch. This leads to inefficiencies, increased handling time, and potential errors.

3. With ASRS:

With ASRS system implementation:

- **Cost Savings:** Eliminating leased storage space will save Rs. 20 crore per year on the rental and maintenance payment to Radiant warehouse and Kothur warehouse, resulting in long-term financial benefits. The pallet storage can be charged at **Rs. 58.95/pallet/day** to the company utilising the ASRS storage.
- **Increased Efficiency:** Automated retrieval and real-time inventory tracking will significantly **reduce material movement time** and improve dispatch accuracy.
- **Enhanced Inventory Control:** Integration with SAP WMS will provide complete **visibility and traceability** of Finished Goods at all times.
- **Improved Order Fulfilment:** Faster and more accurate retrieval of FG pallets will **enhance customer satisfaction** and **reduce return shipments** due to incorrect deliveries as probability of material mismatch is significantly lowered.

With the additional benefit of improved inventory control, better order fulfilment and increased efficiency, a **markup of 15% to the average market cost** of warehouse can be charged to the Lessee.

G. <u>Conclusion:</u>

It is recommended to install the ASRS system under Premier Energies Photovoltaic Gajwel Private Limited (PEPGPL) in the HSTL land. The project shall have a return of **13.10%** with a payback period of **5.16 years** provided the rental cost to the company shall be **Rs. 67.83/ pallet/day.**

Approvers	**Signature**
Sudhir Reddy (Director, Chief Strategy Officer)	
Chiranjeev Saluja (Managing Director)	
Surender Pal Singh (Chairman)	

Annexure 1:

Net Cash Flow (with 15% markup to market cost):

Type	**Cons***	**Cons**	**Cons**	**Cons**	**Cons**	**Cons**	**Cons**	**Cons**	**Cons**	**Cons**	**Cons**	**Cons**	**Cons**	**Cons**
Start Month	**01-Oct-24**	**01-Nov-24**	**01-Dec-24**	**01-Jan-25**	**01-Feb-25**	**01-Mar-25**	**01-Apr-25**	**01-May-25**	**01-Jun-25**	**01-Jul-25**	**01-Aug-25**	**01-Sep-25**	**01-Oct-25**	**01-Nov-25**
End Month	**31-Oct-24**	**30-Nov-24**	**31-Dec-24**	**31-Jan-25**	**28-Feb-25**	**31-Mar-25**	**30-Apr-25**	**31-May-25**	**30-Jun-25**	**31-Jul-25**	**31-Aug-25**	**30-Sep-25**	**31-Oct-25**	**30-Nov-25**
Days	**31.00**	**30**	**31**	**31**	**28**	**31**	**30**	**31**	**30**	**31**	**31**	**30**	**31**	**30**
Count	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Outflows	18.09	0.04	0.04	0.04	0.54	3.84	3.65	3.73	3.73	0.04	0.04	0.04	0.04	3.67
Inflows														
Net Cashflow (I - O)	**(18.09)**	**(0.04)**	**(0.04)**	**(0.04)**	**(0.54)**	**(3.84)**	**(3.65)**	**(3.73)**	**(3.73)**	**(0.04)**	**(0.04)**	**(0.04)**	**(0.04)**	**(3.67)**
Cummulative CF	(18.1)	(18.1)	(18.2)	(18.20)	(18.74)	(22.58)	(26.23)	(29.96)	(33.69)	(33.73)	(33.76)	(33.80)	(33.84)	(37.51)

Type	**Beyond Cons#**	**Beyond Cons**	**Beyond Cons**	**Beyond Cons**	**Beyond Cons**	**Beyond Cons**	**Beyond Cons**	**Beyond Cons**	**Beyond Cons**	**Beyond Cons**	**Beyond Cons**	**Beyond Cons**	**Beyond Cons**	**Beyond Cons**	**Beyond Cons**
Start Month	**01-Dec-25**	**01-Dec-26**	**01-Dec-27**	**01-Dec-28**	**01-Dec-29**	**01-Dec-30**	**01-Dec-31**	**01-Dec-32**	**01-Dec-33**	**01-Dec-34**	**01-Dec-35**	**01-Dec-36**	**01-Dec-37**	**01-Dec-38**	**01-Dec-39**
End Month	**30-Nov-26**	**30-Nov-27**	**30-Nov-28**	**30-Nov-29**	**30-Nov-30**	**30-Nov-31**	**30-Nov-32**	**30-Nov-33**	**30-Nov-34**	**30-Nov-35**	**30-Nov-36**	**30-Nov-37**	**30-Nov-38**	**30-Nov-39**	**30-Nov-40**
Days	**365**	**365**	**366**	**365**	**365**	**365**	**366**	**365**	**365**	**365**	**366**	**365**	**365**	**365**	**366**
Count	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**
Outflows	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Inflows	9.10	9.98	10.54	11.04	11.62	5.88	10.07	9.96	10.88	11.44	5.67	9.16	9.64	10.82	8.36
Net Cashflow (I – O)	**9.10**	**9.98**	**10.54**	**11.04**	**11.62**	**5.88**	**10.07**	**9.96**	**10.88**	**11.44**	**5.67**	**9.16**	**9.64**	**10.82**	**8.36**
Cummulative CF	(28.41)	(18.43)	(7.89)	3.15	14.77	20.65	30.71	40.67	51.55	62.99	68.65	77.82	87.45	98.27	106.63

**Cons = Construction period*

 # Beyond Cons = Beyond Construction period